ASSIGNMENT (Aug 18, 2026)
RNA-Seq Literature Familiarization and Data Characterization

Preparation for RNA-Seq Re-analysis Using Galaxy and GitHub

Part 1. Form Four New Groups 1. Form four new groups

Members	Batigulao, Jehiah Bless Obiñeta, Inah Marie A. Oficiar, Francis Kyle Pasculado, Mark Ryan Suan, Jade Angela
Group Topic	Salinity Stress

Part 2. Find One Published RNA-Seq Study

Article Title	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2
Article Link	https://pmc.ncbi.nlm.nih.gov/articles/PMC10692206/
SAR Database in NCBI	PRJNA943415
SAR Link	https://www.ncbi.nlm.nih.gov/sra/?term=PRJNA943415

Part 3. Understand the Paper Before Downloading Data

As a group, read the Abstract, Introduction, Materials and Methods, and the RNA-seqrelated Results of the paper. Record the following:

Information from the Paper	**Group Answer**
Title of the paper	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in fababean genotype, Hassawi-2
Authors	Muhammad Afzal, Salem S. Alghamdi, Muhammad Altaf Khan, Sulieman A. Al-Faifi, & Muhammad Habib ur Rahman
Year	2023
Journal	*Scientific Reports*, 13, 21223
Organism	Faba bean (*Vicia faba* L.), salt-tolerant genotype Hassawi-2
Tissue / organ / cell type	Leaves at the early vegetative/three-leaf stage

Biological question	To identify genes and molecular mechanisms involved in salinity stress tolerance in Hassawi-2 faba bean at different durations of salt exposure.
Control condition	Faba bean plants grown without the 200 mM NaCl salt treatment
Treatment or stress condition	200 mM NaCl salt stress for 6, 12, 24, 48, and 72 hours
Number of biological replicates	3 biological replicates per condition/time point
Sequencing platform	Illumina NovaSeq 6000 (sequencing outsourced to Macrogen)
Single-end or paired-end sequencing	Paired-end
Read length, if reported	101 bp per read
RNA-seq repository	NCBI Sequence Read Archive (SRA)
BioProject / Study accession	PRJNA943415 (SRA submission ID SUB12947558)
Run accession numbers	Not listed individually in the paper — retrieve the SRR list from the BioProject/SRA page for PRJNA943415
Reference genome or transcriptome used by the authors	None, de novo assembly (no reference genome used; faba bean genome had only just been published)
Main RNA-seq software or tools reported by the authors	Trimmomatic (adapter/quality trimming) → Trinity (de novo assembly) → CD-HIT-EST (clustering into unigenes) → TransDecoder (ORF prediction) → Bowtie (read alignment) → RSEM (expression/FPKM quantification) → edgeR (DEG analysis, TMM normalization) → BLASTX/DIAMOND against NR, Swiss-Prot, KEGG, GO, COG/KOG for annotation

Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram

- **Manipulated:** Salt exposure (200 mM) and duration of stress (6, 12, 24, 48, 72 hours)
- **Control:** Leaf sample at 0 hours (no salt stress)
- **Measured:** Gene expression levels via mRNA transcript sequencing
- **Biological replicates:** 3 biological replicates per condition/time point
- **Tissue:** *Vicia faba* leaf tissue at vegetative stage

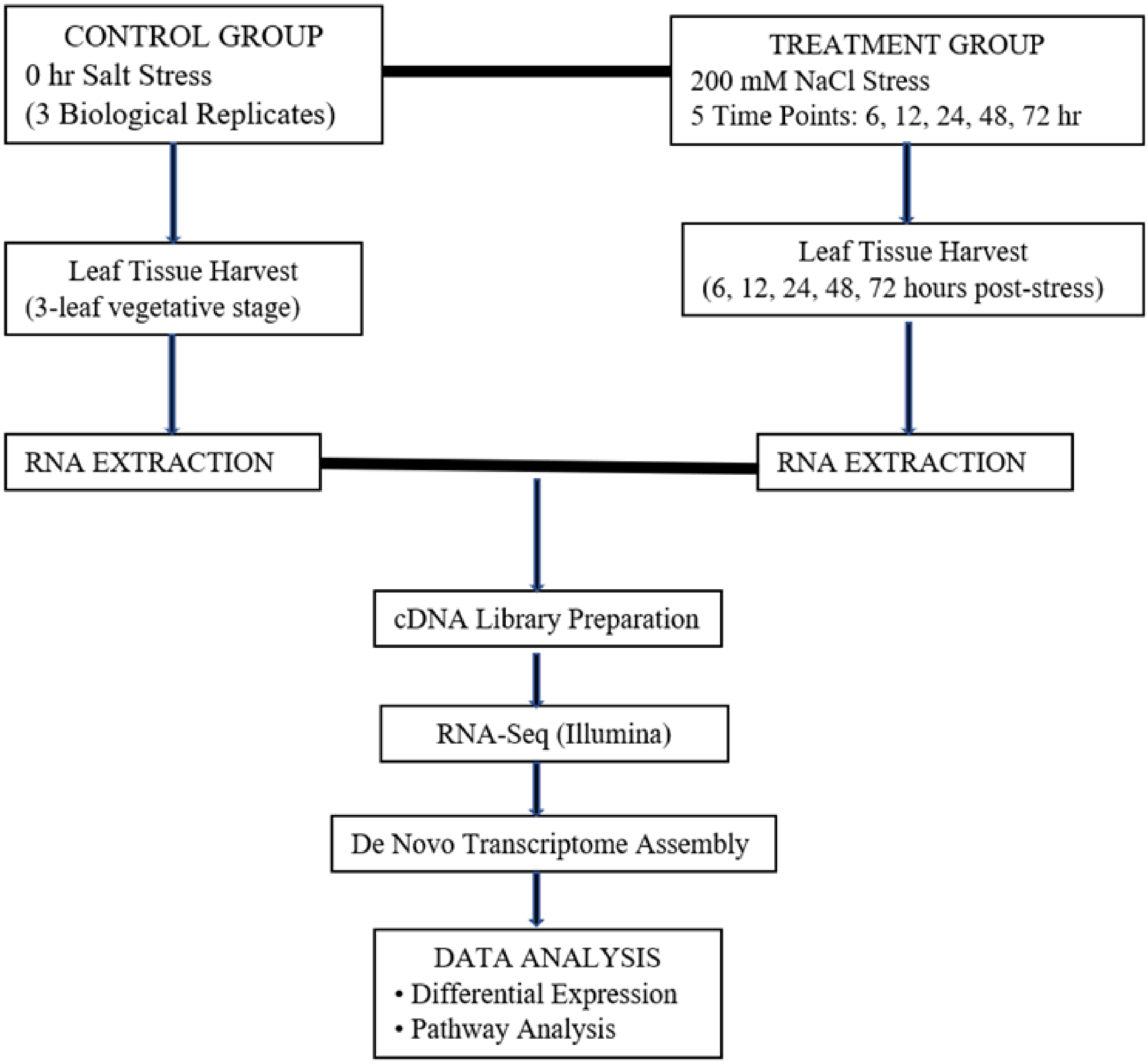

Figure 1. Simple RNA-Seq Experimental Design for Evaluating the Effects of Salt Stress on *Vicia faba* Leaf Gene Expression.

Part 5. Locate the Public RNA-Seq Data

#	Condition	Rep	SRX	Spots	Bases	Pair length (bases/spots)	Per-mate read length
1	Control	R1	SRX19681425	28.2M	5.7G	202.1 bp	101 bp
2	Control	R2	SRX19681431	27.9M	5.6G	200.7 bp	100 bp
3	Control	R3	SRX19681439	28.9M	5.8G	200.7 bp	100 bp
4	Stress 6h	R1	SRX19681440	28.2M	5.7G	202.1 bp	101 bp
5	Stress 6h	R2	SRX19681441	27.7M	5.6G	202.2 bp	101 bp
6	Stress 6h	R3	SRX19681426	25.4M	5.1G	200.8 bp	100 bp
7	Stress 12h	R1	SRX19681427	24.6M	5.0G	203.3 bp	102 bp
8	Stress 12h	R2	SRX19681428	22.4M	4.5G	200.9 bp	100 bp
9	Stress 12h	R3	SRX19681429	28.3M	5.7G	201.4 bp	101 bp
10	Stress 24h	R1	SRX19681430	31.6M	6.4G	202.5 bp	101 bp
11	Stress 24h	R2	SRX19681432	31.5M	6.4G	203.2 bp	102 bp
12	Stress 24h	R3	SRX19681433	26.3M	5.3G	201.5 bp	101 bp
13	Stress 48h	R1	SRX19681434	31.5M	6.4G	203.2 bp	102 bp
14	Stress 48h	R3	SRX19681436	22.5M	4.5G	200.0 bp	100 bp
15	Stress 72h	R1	SRX19681435	31.5M	6.4G	203.2 bp	102 bp
16	Stress 72h	R2	SRX19681437	31.3M	6.3G	201.3 bp	101 bp
17	Stress 72h	R3	SRX19681438	29.5M	6.0G	203.4 bp	102 bp

Part 6. Assign One RNA-Seq Sample to Each Student

Student	Run accession	Condition	Bio. replicate	Single/ Paired-end	Approx. file size
Obiñeta	SRX19681425 (SRR23869388)	Control	Rep 1	Paired-end	1.6 Gb
Suan	SRX19681440 (SRR23869373)	Treatment — 6h salt stress	Rep 1	Paired-end	1.6 Gb
Batigulao	SRX19681430 (SRR23869383)	Treatment — 24h salt stress	Rep 1	Paired-end	1.8Gb
Pasculado	SRX19681434 (SRR23869379)	Treatment — 48h salt stress	Rep 1	Paired-end	1.8 Gb
Oficiar	SRX19681435 (SRR23869378)	Treatment — 72h salt stress	Rep 1	Paired-end	1.8 Gb

Part 7. Download the RNA-Seq Data into Galaxy

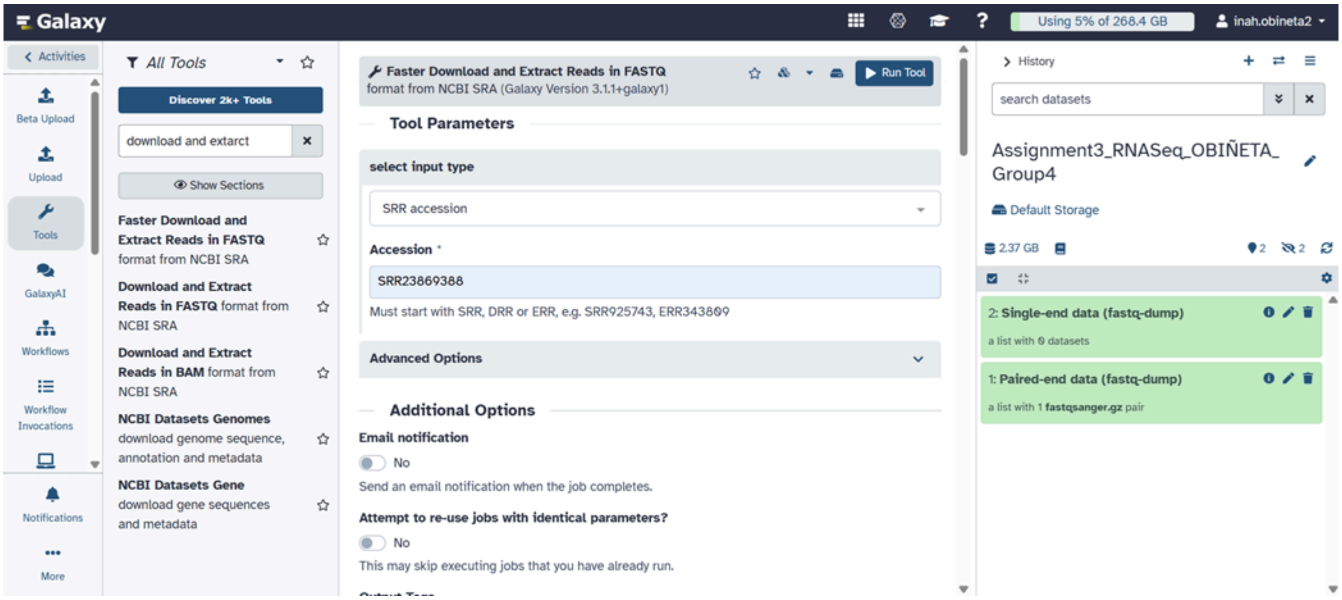

Figure 2. *Successful Download and Extraction of Paired-End and Single-End RNA-Seq Reads in Galaxy using the tool Download and Extract Reads in FASTAQ.* **(OBIÑETA)**

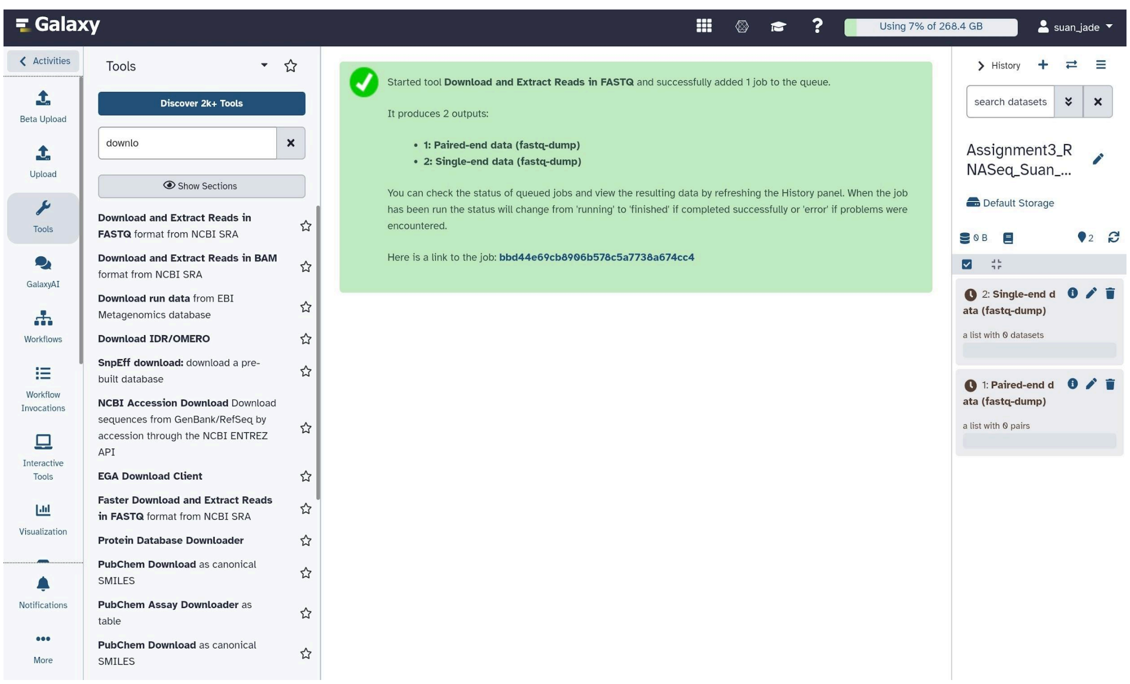

Figure 3. *Galaxy interface showing successful RNA-seq FASTQ import using Download and Extract Reads in FASTQ tool. History named correctly and paired-end output files generated. Student account suan_jade visible.* **(SUAN)**

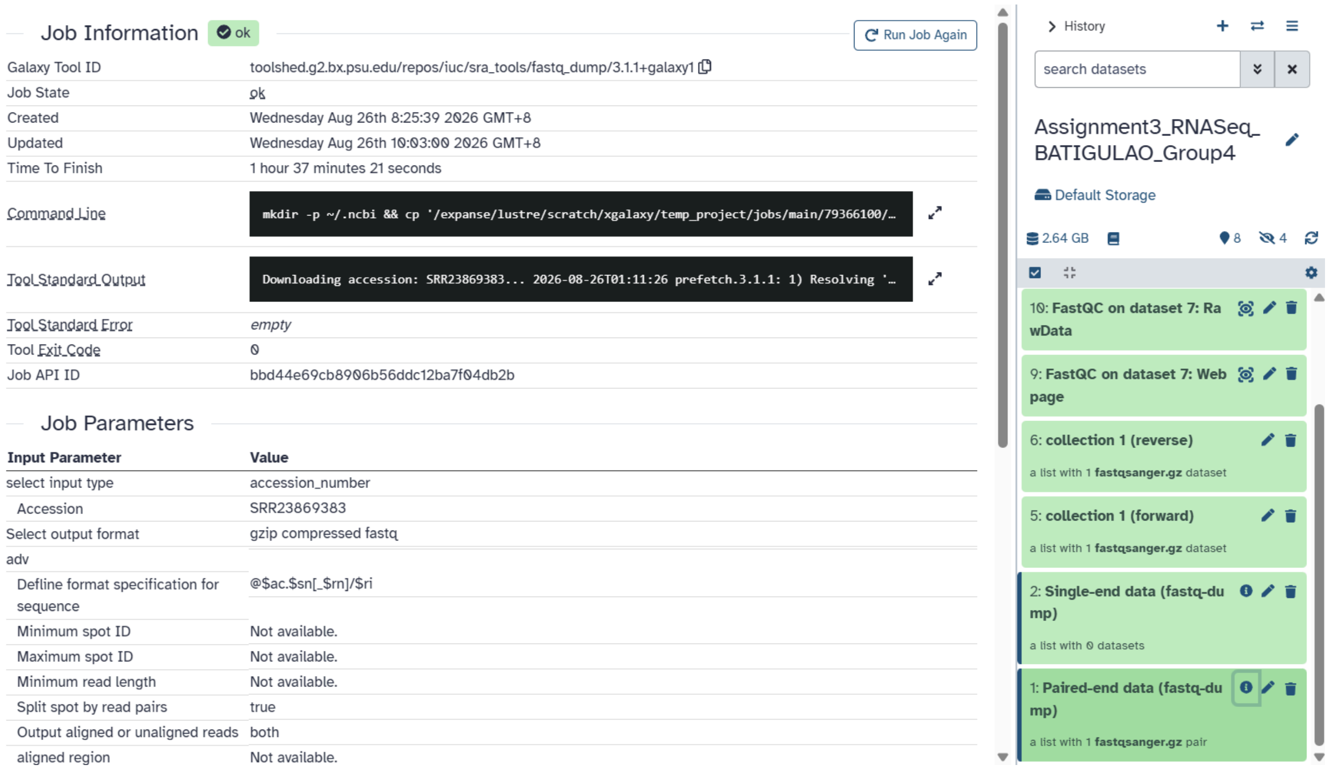

Figure 4. *Successful download and extract of Paired-end and Single-End RNA Seq Reads in Galaxy using the tool download and extract reads in FASTQ format from NCBI SRA.* (**BATIGULAO**)

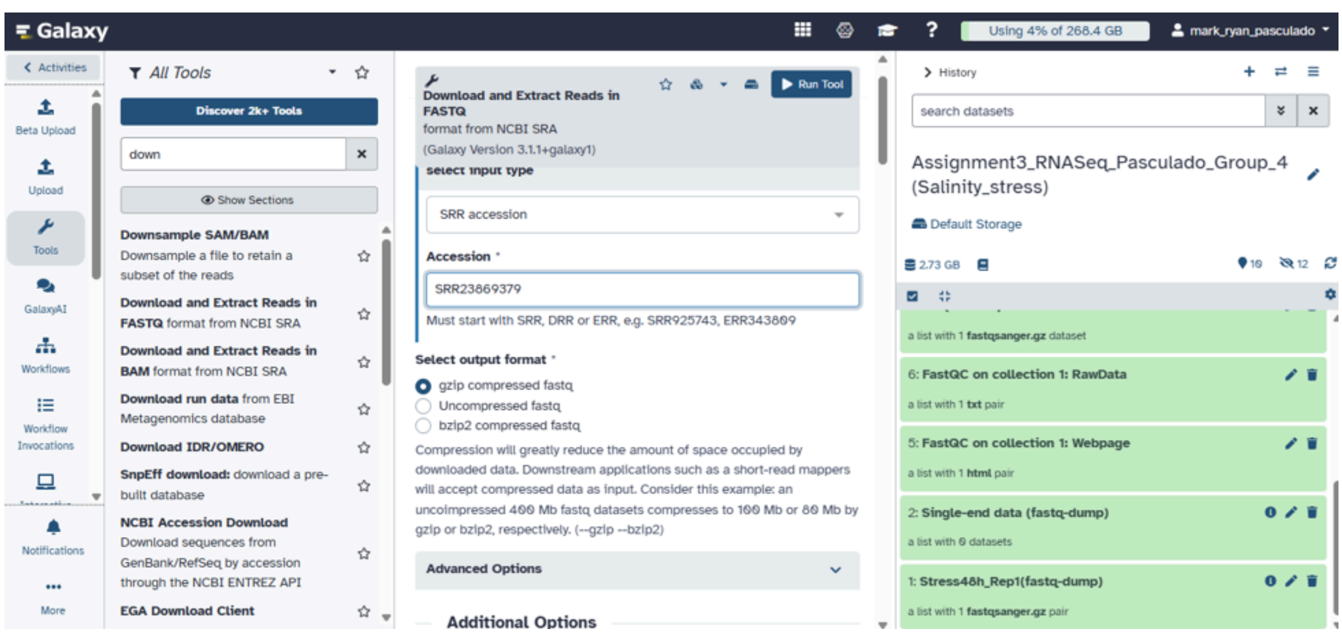

Figure 5. *Galaxy-based download and extraction of paired-end and single-end RNA-Seq reads using the NCBI SRA FASTQ tool.* (**PASCULADO**)

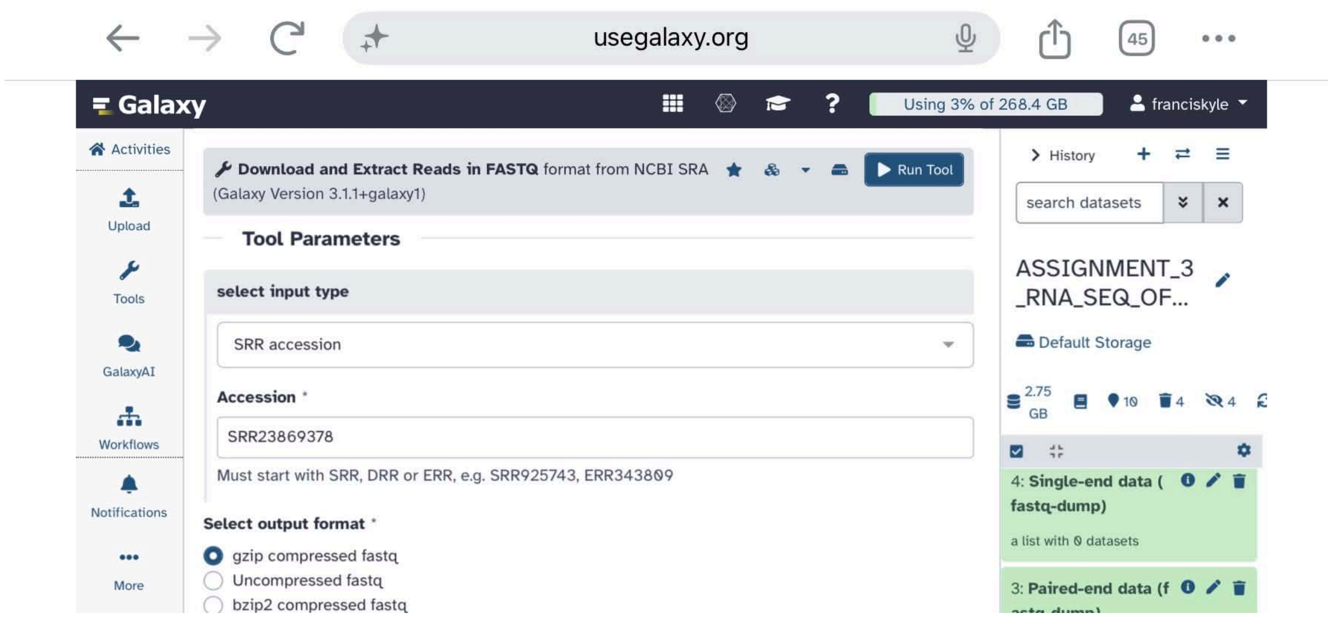

Figure 6. *Configuration of the Download and Extract Reads in FASTQ Tool in Galaxy Using the NCBI SRA Accession SRR23869378 and Selecting Compressed FASTQ Output.* **(OFICIAR)**

Part 8. Examine the FASTQ File

Student	Matrix	Control_Rep1_R1	Control_Rep1_R2
Obiñeta	Line 1: sequence identifier	@SRX19681425.1/1	@SRX19681425.1/2
	Line 2: nucleotide sequence	NCTGCATCTTCTAATC ACCACCGGCATGCTT GAACTCTTGAAGTTC TTTAAAATTCAGCCA TGGTTTGACCCATAC TTGAGCTTCATAGAT CTTCTGCCCT	GCGCCTGCTGCCTTCCTTGG ATGTGGTAGCCGTGTCTCAG GCTCCCTCTCCGGAATCGAA CCCTAATTCTCCGTCACCCGT CACCACCATGGTCGGCCACT
	Line 3: separator line	+	+
	Line 4: quality-score characters	#FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFF FFFFFFFF,FFFFFFFFF FFF:FFFFFFFFFFFFFF FFFFFFFFFFF:FFF:F:F FFFFFFF	FFFFFFFFFFFFFFF,FFFF:FFFF F,FFFFF,FFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFF:FFFF,FFFFFFF
		Treatment6h_Rep1_R1	**Treatmen6ht_Rep1_R2**

Suan	Line 1: sequence identifier	@SRR2386973.1/1	@SRR23869373.1/2
	Line 2: nucleotide sequence	NCCCGACCCGACCCATGTTTTTGAGTTGGTGTATTTGGAAGACTACAAAACCCGGGTCGAAGCACAGAACATCTCGGTGACCCGAAACTTGACCCGAATGT	GTAAATCTTTAGGTTCATCTTTGGGTTCAAGTTCAGTTAACGAGATGGTTGCTTCCTTAAGAAATTTACAGTTAGGGAAGTTAAAATCTATGCCGGTTAAT
	Line 3: separator line	+	+
	Line 4: quality-score characters	#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FF,	FFFF,FFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFF:FFFF,FFFFFFFFFFFFF::FFFFFF,FFFFFFFF
		Treatment24_Rep1_R1	**Treatmen24t_Rep1_R2**
Batigulao	Line 1: sequence identifier	@SRR23869383.1/1	@SRR23869383.1/2
	Line 2: nucleotide sequence	NCGGATTCTGACTTAGAGGCGTTCAGTCATAATCCAACGCACGGTAGCTTCGCGCCACTGGCTTTTCAACCAAGCGCGATGACCAATTGTGCGAATCAACG	GATGACAGTGTCGCAATAGTAATTCAACCTAGTACGAGAGGAACCGTTGATTCGCACAATTGGTCATCGCGCTTGGTTGAAAAGCCAGTGGCGCGAAGCTA
	Line 3: separator line	+	+
	Line 4: quality-score characters	#FFFFFFFFFFFFFFFFFFFFF::F:FF:FFFFFFFFFFFFFFFFFFF:F,FFFFFFF:FFF:FFFFFFFFF,:FFFF:FF:FF:F:FFFFFFFF:FFFFF	FFFFFFFFFFFFFF,:FFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFF:FFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFF
		Treatment48_Rep1_R1	**Treatment48_Rep1_R2**

Pasculado	Line 1: sequence identifier	@SRR23869379.1/1	@SRR23869379.1/2
	Line 2: nucleotide sequence	NCCATGTTGAGCTCC TCAAATCTGGCACGA GTAATGTGGGAATAG AAATCAATTCCCTCA TATAGAGAATCAATC TCAATGGTTGTCTGG GCAGTGGAAGA	GGTACTTTTGATGTCTCTTT GCTTACTATTGAAGAGGGTA TCTTTGAGGTCAAAGCTACA TCTGGAGACACCCATCTTGG AGGGGAGGATTTTGACAAC AG
	Line 3: separator line	+	+
	Line 4: quality-score characters	#,,FFFFFFFFF:FF:FFFF FFFFFFF,:FFFFFF,,F:,F FFFF:,F:FFF:F:FFFFFF: ,FF:FF,FFF,F:FFFFFFF FF,FFFF:FFFFFFFFFF	F,:,FF,,,,:FFFFFFFFFF,,:FFFF::F, FF::,,:,F::F,,F:FFFFFF,FF:,F,FF,,, FF::F,:FF::FF:FF,,:F,FFFFFFFF,, FFF
		Treatment72_Rep1_R1	**Treatment72_Rep1_R2**
Oficiar	Line 1: sequence identifier	@SRR23869378.1/1	@SRR23869378.1/2
	Line 2: nucleotide sequence	NTAATGTGACAGT GGTGTTACATTTG GTAATTTAGGACC AGGAAATTAAGTC TGTCCCAGGGCA ATTAGGAACCCAA TACACGAGTTCTG AAACATTAAAG	CTGGAGGATGGAAGGAC CCTTGCTGACTACAACA TTCAGAAGGAGTCTACT CTTCACCTTGTATTGCGC CTTCGTGGTGGATTTTAG CTTTTTTCATGTTT
	Line 3: separator line	+	+
	Line 4: quality-score characters	#FFF:FF:FFFFFFFFF FFF:,FFFFFFFFFFF FFFFF,FFFFFFFFFF	FFFFF::,FF:FFFFFFFFFFF FFFF:FFFFFF:FFFFFFFFF FFFFFFFFFFFFFFFFFFFFF

		FFF::FFFF:F:FF,FFF FFFFFFFFFFFFFF FF:,FFFFFFFFFFFF F	:FFFFFFFFF:FFF:FFFFFF FFFFFFF,,FF,F

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?
   - FASTA files only have the sequence name and the nucleotide sequence itself, just two lines per entry. FASTQ files have four lines for each read: the sequence identifier, the nucleotide sequence, a separator line, and then a quality-score character. So unlike FASTA, FASTQ includes quality information along with the sequence.

2. What information is present in FASTQ that is not present in FASTA?
   - FASTQ has per-base quality scores that FASTA doesn't have. These are characters that represent how accurate each base call is, they tell you the probability that the nucleotide was sequenced correctly. FASTA only shows the sequence and nothing about its quality.

3. Why are quality scores important in RNA-seq?
   - Quality scores are important because they tell us how reliable the sequencing data is. If a base has a low quality score, there's a higher chance it was called incorrectly. Bad quality reads can mess up all the downstream analysis, like alignment and gene expression counts. Checking the quality helps us know if we can trust the data, and also tells us which parts might need to be trimmed or filtered out before we do any further analysis.

Part 9. Run FastQC

Figure 7. *Galaxy History Panel Showing the RNA-Seq Analysis Workflow and Generated FastQC Quality Control Outputs of both forward and reverse* **(OBIÑETA)**

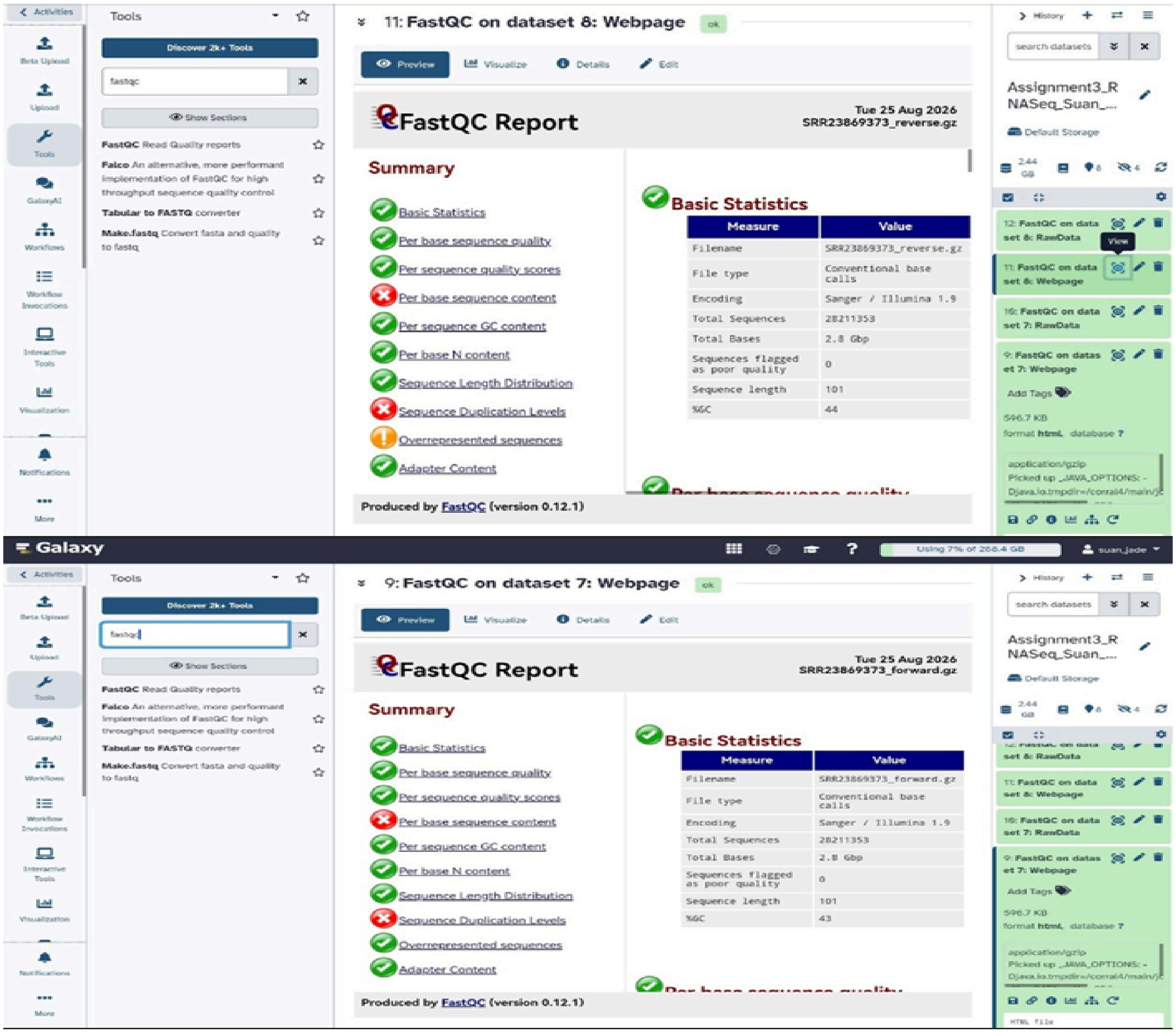

Figure 8. *FastQC Reports — Basic Statistics Summary for Forward (R1) and Reverse (R2) Reads* **(SUAN)**

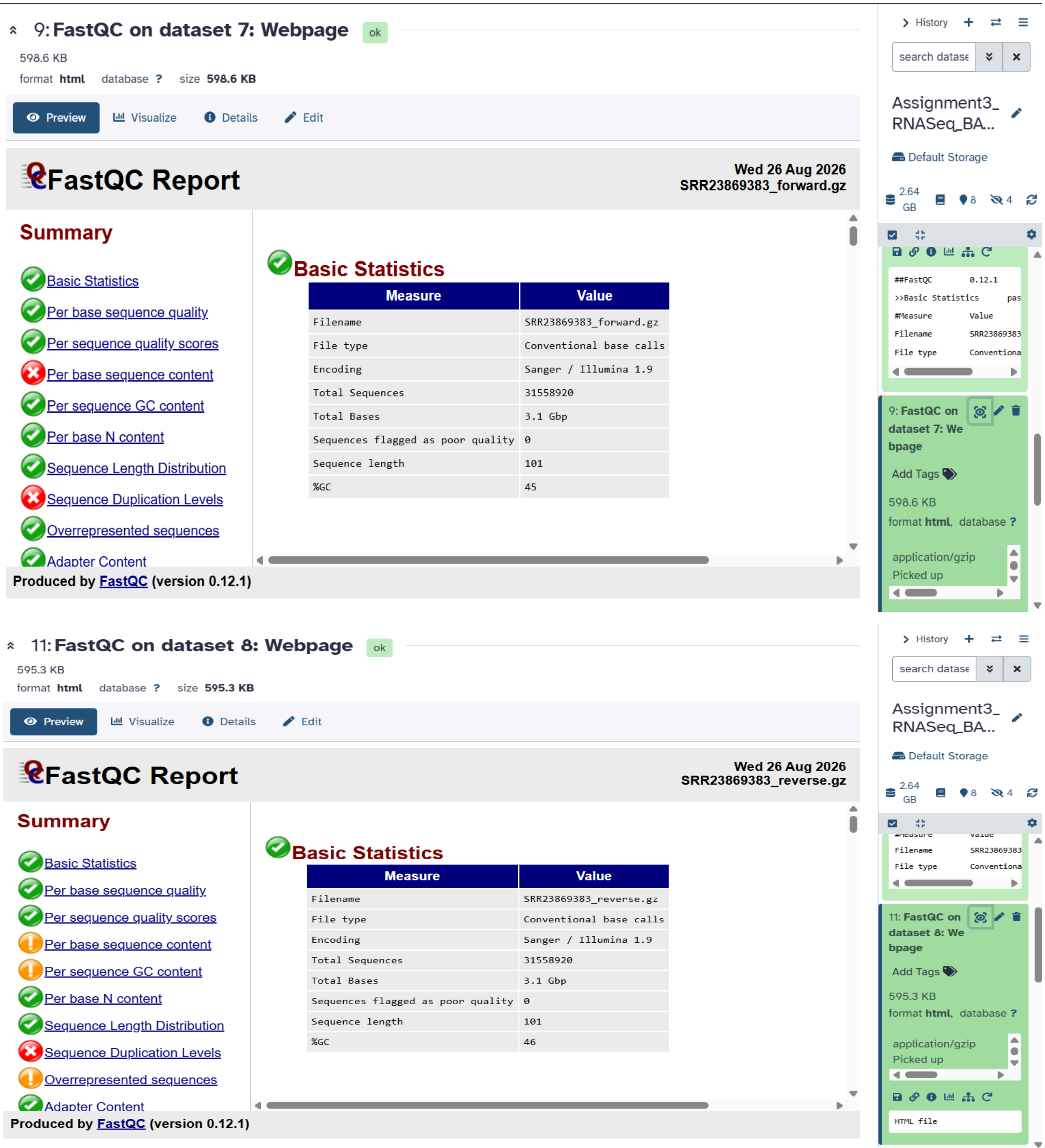

Figure 9. *FastQC Reports — Basic Statistics Summary for Forward (R1) and Reverse (R2) Datasets* **(BATIGULAO)**

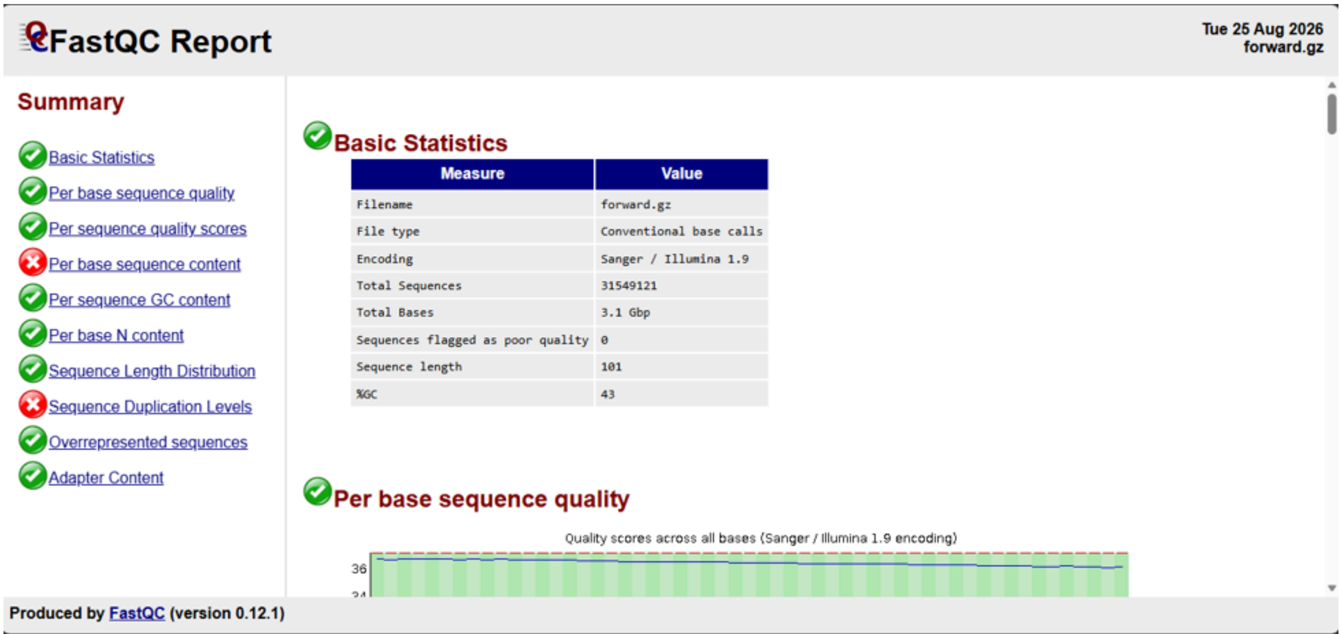

Figure 10. *FastQC Report on forward read (R1) RNA-seq FASTQ file. (**PASCULADO**)*

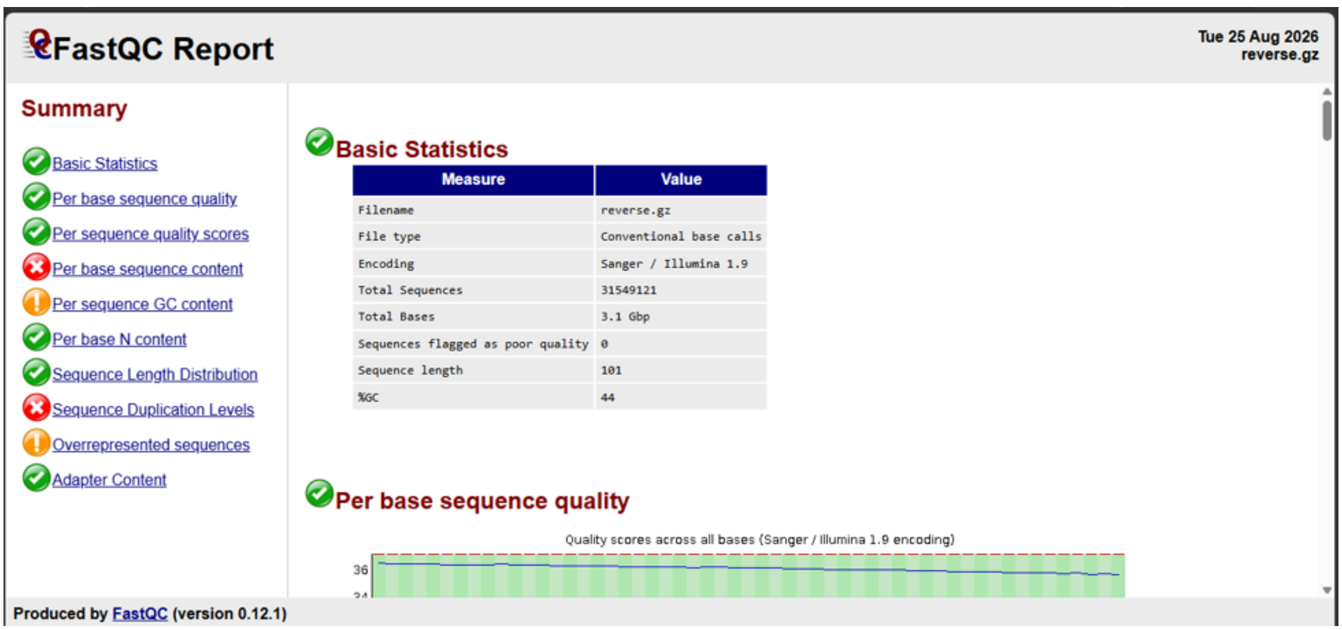

Figure 11. *FastQC Report on reverse read (R2) RNA-seq FASTQ file (**PASCULADO**)*

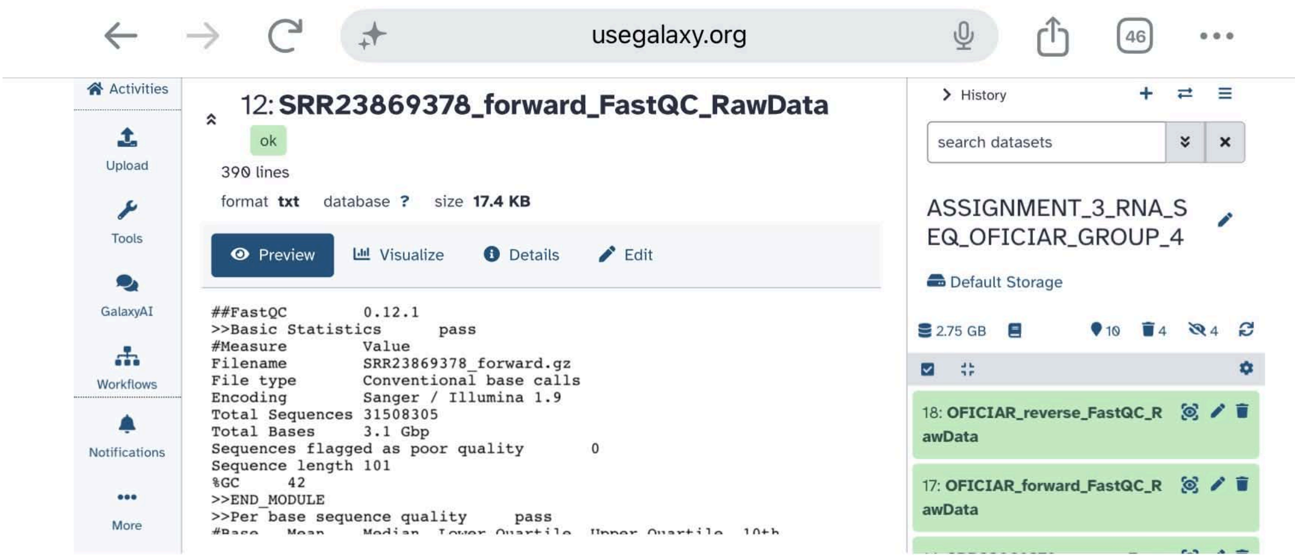

Figure 12. FastQC Report on forward read (R1) RNA-seq FASTQ file. (**OFICIAR**)

Figure 13. FastQC Report on reverse read (R2) RNA-seq FASTQ file. (**OFICIAR**)

Part 10. Characterize Your RNA-Seq Dataset

OBIÑETA: Control; Replicate 1

RNA-Seq Characteristic	**Your Observation**
RNA-seq run accession	RI&R2: SRX19681425
Condition	RI&R2: Control Replicate 1
Single-end or paired-end	RI&R2: Paired-end
Number of reads / sequences	RI&R2: 28,209,473
Read length	RI&R2: 101 bp
File size	RI&R2: 1.2 GB
GC content	RI&R2: 45%
General per-base sequence quality	RI:36.5 R2:36.6
Presence of adapter content	RI&R2: Pass — no significant adapter contamination detected in either
Presence of overrepresented sequences	RI&R2: Warning — some overrepresented sequences flagged
Any quality warning observed	R1: FAIL — Per base sequence content; FAIL — Sequence Duplication Levels; WARN — Overrepresented sequences. R2: same as R1, plus WARN — Per sequence GC content

SUAN: Treatment_6hrs; Replicate 1

RNA-Seq Characteristic	**Your Observation**
RNA-seq run accession	SRR23869373
Condition	Treatment — 6 h salt stress (200 mM NaCl), Replicate 1
Single-end or paired-end	Paired-end
Number of reads / sequences	28211353
Read length	101 bp
File size	R1 and R2: 1.2 GB
GC content	R1: 43% / R2: 44%
General per-base sequence quality	R1: Q ~36, all green / R2: Q ~35–36, all green

Presence of adapter content	R1 and R2: 0% all adapters
Presence of overrepresented sequences	R1: None found/ R2: PolyG artifact (0.24%), Illumina-specific, not contamination
Any quality warning observed	R1: No warnings. Red flags on sequence content & duplication are normal for RNA-seq/ R2: Minor warning — poly-G artifact (0.24%), common sequencing noise, negligible level. All critical metrics pass

BATIGULAO: Treatment_24hrs; Replicate 1

RNA-Seq Characteristic	**Your Observation**
RNA-seq run accession	R1&R2: SRR23869383
Condition	R1&R2: Treatment 24 hours
Single-end or paired-end	R1&R2: Paired-end
Number of reads / sequences	R1&R2: 31558920
Read length	R1&R2: 101bp
File size	R1&R2: 1.8Gb
GC content	R1: 45% & R2: 46%
General per-base sequence quality	R1&R2: pass
Presence of adapter content	R1&R2: pass
Presence of overrepresented sequences	R1: pass & R2: Warning — poly-G sequence detected (0.48%)
Any quality warning observed	Yes, R1 showed failures for per-base sequence content and sequence duplication levels. R2 showed warnings for per-base sequence content, per-sequence GC content, and overrepresented sequences, and failed sequence duplication levels.

PASCULADO: Treatment_48hrs; Replicate 1

RNA-Seq Characteristic	**Your Observation (Forward)**	**Your Observation (Reverse)**
RNA-seq run accession	SRR23869379	SRR23869379
Condition	R1 (Forward read)	R2 (Reverse read)
Single-end or paired-end	Pair-end	Pair-end

Number of reads / sequences	31,549,121	31,549,121
Read length	~98 bp	~98 bp
File size	1.8GB	1.8GB
GC content	Pass	Warning
General per-base sequence quality	Pass	Pass
Presence of adapter content	Pass	Pass
Presence of overrepresented sequences	Pass	Warning
Any quality warning observed	Fail: Per base sequence content	Warn: Per sequence GC content Fail: Per base sequence content

OFICIAR: Treatment_72hrs; Replicate 1

RNA-Seq Characteristic	Your Observation (Forward)	Your Observation (Reverse)
RNA-seq run accession	SRR23869379	SRR23869379
Condition	R1 (Forward read)	R2 (Reverse read)
Single-end or paired-end	Pair-end	Pair-end
Number of reads / sequences	31508305	31508305
Read length	~101 bp	~101 bp
File size	1.8GB	1.8GB
GC content	Pass	Warning
General per-base sequence quality	Pass	Pass
Presence of adapter content	Pass	Pass

Presence of overrepresented sequences	Pass	Warning
Any quality warning observed	Fail: Per base sequence content; Sequence Duplication Levels.	Warn: Per sequence GC content ; Overrepresented sequences.

Part 11. Group MultiQC Summary

MultiQC Observations for Forward and Reverse Reads					
Comparison Point	**Obiñeta (SRR23869388)**	**Suan (SRR23869373)**	**Batigulao (SRR23869383)**	**Pasculado (SRR23869379)**	**Oficiar (SRR23869378)**
Condition	Control, Rep 1	Treatment — 6h salt stress	Treatment — 24h salt stress	Treatment — 48h salt stress	Treatment — 72h salt stress
Number of reads	28.21M	28.21M	31.56M	31.55M	31.51M
Read quality	High — 20% modules flagged	High — 20% modules flagged	High —10–20% modules flagged (best of group)	High — 20% modules flagged	High — 20% modules flagged
GC content	45%	43–44%	45–46%	43–44%	42–43%
Adapter contamination	None detected	None detected	None detected	None detected	None detected
% Duplicates	77–79%	70–72%	77–79% (highest)	74–76%	69–71%
Notes / outlier flag	Typical, in line with group average	Typical, in line with group average	Slight outlier — lowest fail rate, highest GC%, highest duplication	Typical, in line with group average	Typical, in line with group average

Part 12. Compare the RNA-Seq Samples Within the Group

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Obiñeta	SRR23869388	Control — 0h salt stress	28,209,473	101 bp	45%	Good quality, adapter-trimmed, uniform base composition; no major bias
Suan	SRR23869373	Treatment — 6h salt stress	28,211,353	101 bp	43-44%	Comparable to control; R2 has minor poly-G artifact (0.24%); all critical metrics pass
Batigulao	SRR23869383	Treatment — 24h salt stress	31,558,920	101 bp	45-46%	Main QC Observation: Approximately 45–46% of the reads passed the quality control, indicating moderate read quality. Further trimming and filtering may be needed before downstream analysis
Pasculado	SRR23869379	Treatment — 48h salt stress	31,549,121	101 bp	43-44%	Good overall quality with 31.5M reads and 43–44% GC, but FastQC showed per-base sequence content failure and a GC-content warning in R2
Oficiar	SRR23869378	Treatment — 72h salt stress	31,508,305	101 bp	43%	High duplication and sequence bias typical for RNA-seq, with significant Poly-G stretches in reverse reads caused by Illumina two-color chemistry.

Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?
   - The study aimed to find out which genes and biological pathways help faba bean (genotype Hassawi-2) tolerate salt stress. By comparing gene expression in normal conditions versus salt stress over time (6 to 72 hours), the researchers wanted to understand how the plant responds to salinity at the molecular level and identify the specific genes that allow it to grow under high-salt conditions.
2. Why did the authors use RNA-seq instead of only examining the genome?
   - The genome contains all genes, but it does not show which genes are actually active or turned on at a given time or condition. RNA-seq measures gene expression, it tells us which genes are being transcribed into mRNA and at what levels. This is important because salinity stress changes how genes are expressed, and those changes cannot be seen by just looking at the DNA sequence.
3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?
   - Genomic DNA contains the full set of genetic information in every cell and stays the same throughout the plant's life and in all conditions. RNA, on the other hand, is the transcribed copy of active genes. RNA-seq measures mRNA, which represents the genes that are currently being expressed, whereas DNA represents all genes whether active or not.
4. What is a biological replicate and why is it important?
   - A biological replicate is an independent sample in this study, it means leaves from a separate plant grown and treated under exactly the same conditions. Biological replicates are important because they account for natural variation between individual plants. They help confirm that observed differences are truly caused by salinity stress and not just by chance or variation in one particular plant. They also make the results statistically reliable.
5. What is the difference between single-end and paired-end sequencing?
   - In single-end sequencing, we read only from one end of the DNA fragment, producing one sequence per fragment. In paired-end sequencing, we read from both ends the forward (R1) and reverse (R2), producing two sequences that point toward each other. Paired-end reads provide more information, improve alignment accuracy, and help detect structural variations, which is why it was used in this study.
6. What is a FASTQ file?
   - A FASTQ file is the standard output format from high-throughput sequencing. It stores both the nucleotide sequence (the actual A, T, C, G bases) and a quality score for each base, indicating how confident the sequencer is that the base was called correctly. Each sequence entry has four lines: an identifier, the sequence itself, a separator line, and the quality scores encoded as characters.
7. What information does FastQC provide?
   - FastQC is a quality control tool that generates a comprehensive report to assess the quality of raw sequencing data. It provides statistics such as total number of reads, read length, GC content, per-base quality scores, sequence duplication levels, presence of adapters, and overrepresented sequences. It helps us check if the data is clean, reliable, and suitable for analysis.

8. What does a high per-base quality score indicate?
    - A high per-base quality score means the base call at that position is highly accurate and reliable. Quality scores are on a logarithmic scale: Q30 means a 1 in 1,000 chance of an incorrect base call. High scores across all positions indicate good sequencing quality and give confidence that the sequence data is trustworthy.
9. Why can adapter contamination be a problem?
    - Adapter sequences are short DNA fragments attached to the ends of the DNA during library preparation. If they are not completely removed before sequencing, they may appear in the reads. This is a problem because adapters are not part of the organism's genome, if left in the data, they interfere with read alignment, cause mismatches, and can lead to incorrect or unreliable results during downstream analysis.
10. Were all RNA-seq samples in your group similar in quality? Explain.
    - Yes, all samples showed very similar and excellent quality. Every sample had consistently high per-base quality scores (approximately Q35–Q36) across the entire read length, uniform read length of 101 bp, no adapter contamination, and similar GC content around 43–45%. Read counts were also comparable at approximately 28–31 million reads per sample. The only minor difference was one R2 file showing a very low level of poly-G sequence, which is a harmless sequencing artifact and not a true quality issue.
11. Did any sample show a possible quality problem? What was it?
    - One R2 file showed a yellow warning flag for overrepresented sequences, a short poly-G sequence appeared at a very low level of only 0.24%. This is not a real problem, it is a well-known artifact specific to Illumina sequencing caused by signal decay in the G nucleotide channel, and it is not biological contamination. The percentage is extremely low and does not affect the reliability of the data.
12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

    First, trim low-quality bases and remove adapters. Then, map clean reads to a reference or assemble a transcriptome. Next, count reads per gene, normalize for differences in sample size, and use statistical tests to find genes with significant expression changes between groups.

Part 14. GitHub Documentation

GitHub Group Repository Link:
https://github.com/ryanpas123e/Group4_RNASeq_Familiarization.git

Part 15. Group Summary

1. What study did the group choose and why is it relevant to the assigned topic?
    - Our group selected "Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2" by Afzal et al. (2023, *Scientific Reports*). This study directly matches our assigned topic of salinity stress, since it examines how a plant's gene expression shifts in response to salt exposure over time, comparing an unstressed control group to salt-treated samples at five different time points

2. What organism, tissue, control, and treatment were used?
    - The organism is faba bean (*Vicia faba* L.), specifically the salt-tolerant genotype "Hassawi-2." The tissue sampled was leaf tissue from seedlings at the early vegetative (three-leaf) stage. The control group consisted of untreated seedlings, while the treatment group was exposed to 200 mM NaCl and sampled at five time points after treatment: 6, 12, 24, 48, and 72 hours.
3. How many RNA-seq samples were included in the original study?
    - The study included 18 total samples with six conditions (control plus five stress time points), each with three biological replicates. On NCBI SRA, however, only 17 of these runs are currently listed; one replicate (Stress 48h, Rep 2) does not appear to be publicly available.
4. What sequencing platform and read type were used?
    - The authors used Illumina paired-end sequencing (HiSeq 4000), with a read length of 101 bp per read.
5. What did the group learn from examining the raw RNA-seq data?
    - Working directly with the FASTQ files and FastQC reports taught us how to interpret real sequencing quality metrics rather than just reading about them. We experienced firsthand that RNA-seq data has some "expected" quirks like biased base composition at the start of reads and high sequence duplication that would look alarming in genomic DNA sequencing but are normal and expected for RNA-seq, since highly expressed genes naturally produce many duplicate reads. We also learned how much manual effort goes into consolidating data across a group when everyone works in separate accounts, since Galaxy doesn't support real-time shared histories the way we initially assumed.
6. Were there noticeable differences in data quality among samples?
    - Overall, all five group members' samples were very consistent in quality, similar read counts (roughly 28–32 million reads per file), similar GC content (42–46%), and no adapter contamination in any sample. One sample, Batigulao's (SRR23869383, 24h stress), stood out slightly with a marginally lower FastQC fail rate and the highest GC content and duplication rate in the group, but this was a minor difference rather than a quality concern.
7. What analysis did the original authors perform after quality control?
    - After quality filtering and trimming (using Trimmomatic), the authors performed de novo transcript assembly with Trinity, since no reference genome was used. They then clustered and annotated the resulting transcripts using tools like CD-HIT-EST, TransDecoder, and BLAST-based searches against databases such as NR, Swiss-Prot, KEGG, and GO. Differential expression between control and each stress time point was calculated using edgeR, and selected results were validated experimentally with qRT-PCR.
8. What will the group need to learn next in order to reproduce part of the authors' gene expression analysis?
    - Since our group only has computers and Galaxy, all next steps stay fully computational, just like this assignment. We'll need to learn how to trim reads (e.g., with Trimmomatic), then either align to a reference genome or perform de novo assembly with Trinity, as the original authors did. From there, we'll need to generate expression counts and use a tool like edgeR to statistically compare control and treatment samples. All of this happens through Galaxy's web-based tools, the same way we ran FastQC here where no wet-lab work required.